

Solution Architecture

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Team ID	SWTID-2026-9792
Project Name	FitBuddy – AI Fitness Plan Generator

Solution Architecture

Presentation / Client Layer

HTML + CSS + Jinja2 templates: index.html, result.html, feedback.html, admin_login.html and all_users.html.



Application Layer – FastAPI

FastAPI routes handle form submission, workout generation, feedback, admin login, dashboard access and health checks.



AI Integration Layer

Google Gemini API using gemini-3.5-flash-lite for workout generation, nutrition/recovery tips and plan updates.



Data / Storage Layer

SQLite database accessed through SQLAlchemy models User and Plan.

Component Description Table

Component	Description / Role	Technology
Presentation Layer	Collects user input and displays plans, feedback and admin data.	HTML, CSS, Jinja2
Application Layer	Routes requests, validates form data and coordinates services.	Python, FastAPI
AI Service	Generates and revises fitness content.	Google Gemini API
Database	Stores users, original plans, updated plans, nutrition tips and feedback.	SQLite, SQLAlchemy
Authentication	Checks admin credentials and maintains login session.	FastAPI SessionMiddleware + .env